

HYPOTHESIS TESTING

Power and the Neyman-Pearson Lemma

Review/setting the stage

- Hypothesis (H): A statement about a probability distribution function (usually in terms of the distribution's parameters)
- Null hypothesis (H_0): “Default”: Assume H_0 is true until the data “disprove” it (“disprove” H_0 means “reject H_0 ”).
- Alternative hypothesis (H_a): When we reject H_0 we shift belief to H_a .
- $\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$
- $\beta = P(\text{Type II error}) = P(\text{Accept } H_0 \mid H_a \text{ is true})$
- $\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_1 \text{ is true})$
- Typically, we set α to some “acceptable small” value (like 0.05) before observing data.

Simple and composite hypotheses

- Definition: A simple hypothesis uniquely specifies the distribution of the data.

- Examples:

- For Poisson(λ) data, $H: \lambda = 4$

Poisson(4)

- For binomial(n, p) data (n set by design), $H: p = \frac{1}{3}$

Bi($n, \frac{1}{3}$)

- For normal($\mu, 1$) data, $H: \mu = 100$

N(100, 1)

- Definition: A composite hypothesis is any other hypothesis

- Examples:

- For Poisson(λ) data, $H: \lambda < 4$

Poisson(3.99)

Poisson(2)

- For binomial(n, p) data (n set by design), $H: p \neq \frac{1}{3}$

- For normal(μ, σ^2) data, $H: \mu = 100$ (it specifies μ but σ^2 could be anything > 0)

N(100, 7)

N(100, 15, 2)

Testing strategy for two simple hypotheses

- Choose α
- Given α , determine rejection region to maximize power / minimize β .
- Neyman-Pearson Lemma:

To test simple hypotheses $H_0: \theta = \theta_0$ and $H_a: \theta = \theta_a$ based on iid random variables $Y_1, Y_2, \dots, Y_n$, for any fixed α , the most powerful test has a rejection region defined by

$$\frac{L(Y_1, Y_2, \dots, Y_n | \theta_0)}{L(Y_1, Y_2, \dots, Y_n | \theta_a)} \leq k$$

where k is chosen to ensure $P(\text{Type I error}) = \alpha$.

Likelihood function

Reject H_0 if
$$\frac{L(\theta_0)}{L(\theta_a)} \leq k$$

$$f(y) = \frac{1}{\theta} e^{-\frac{y}{\theta}}, y \geq 0$$

$$F(y) = \begin{cases} 1 - e^{-\frac{y}{\theta}}, & y \geq 0 \\ 0 & y < 0 \end{cases}$$

$$L(\theta) = L(Y|\theta) = \frac{1}{\theta} e^{-\frac{y}{\theta}}$$

Neyman-Pearson example: One observation

Based on one observation $Y \sim \text{exponential}(\theta)$ and hypotheses are $H_0: \theta = 2; H_a: \theta = 1$.

a. Find the most powerful test with $\alpha = 0.05$.

b. Calculate the power of the test.

(a) Reject H_0 if $\frac{L(Y|\theta_a)}{L(Y|\theta_0)} \leq k$

$$\text{if } \frac{\frac{1}{2} e^{-\frac{y}{2}}}{1 \cdot e^{-y}} \leq k$$

$$\text{if } \frac{1}{2} e^{\frac{y}{2}} \leq k$$

$$\text{if } e^{\frac{y}{2}} \leq 2k$$

$$\text{if } \frac{y}{2} \leq \log(2k)$$

$$\text{if } y \leq 2 \log(2k) = c$$

(if Y "small")

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true}) = 0.05$$

$$= P(Y \leq c \mid Y \sim \text{exp}(2)) = 0.05$$

$$1 - e^{-\frac{c}{2}} = 0.05$$

$$-e^{-\frac{c}{2}} = -0.95$$

$$e^{-\frac{c}{2}} = 0,95$$

$$-\frac{c}{2} = \log(0,95)$$

$$c = -2 \log(0,95) \approx 0,1026$$

Reject H_0 if $Y \leq 0,1026$

$$RR: \{Y: Y \leq 0,1026\}$$

$$(b) \text{ power} = P(\text{reject } H_0 \mid H_a \text{ is true})$$

$$= P(Y \leq 0,1026 \mid \theta = 1)$$

$$= 1 - e^{-0,1026/1}$$

exp(1)

$$= 0,0975$$

$$f(y) = \frac{1}{\sqrt{2\pi} \cdot 6} e^{-\frac{(y-\mu)^2}{2 \cdot 36}}$$

$$L(Y_1, \dots, Y_n | \mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \cdot 6} e^{-\frac{(Y_i - \mu)^2}{2 \cdot 36}} = \left(\frac{1}{\sqrt{2\pi} \cdot 6} \right)^n e^{-\frac{1}{72} \sum (Y_i - \mu)^2}$$

Another Neyman-Pearson example: Random sample

Observations $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu, 36)$.

a. Find the most powerful test of $H_0: \mu = 50$ vs. $H_1: \mu = 55$.

b. Calculate the power of the test when $n = 9$.

c) Reject H_0 if $\frac{\left(\frac{1}{\sqrt{2\pi} \cdot 6}\right)^n e^{-\frac{1}{72} \sum (Y_i - 50)^2}}{\left(\frac{1}{\sqrt{2\pi} \cdot 6}\right)^n e^{-\frac{1}{72} \sum (Y_i - 55)^2}} \leq k$

if $\exp\left\{-\frac{1}{72} \left[\sum Y_i^2 - 2 \cdot 50 \sum Y_i + n 50^2 - \left(\sum Y_i^2 - 2 \cdot 55 \sum Y_i + n 55^2 \right) \right]\right\} \leq k$

$n \bar{Y} = \sum Y_i$

if $-\frac{1}{72} \left[\cancel{\sum Y_i^2} - \underline{100n \bar{Y}} + n 50^2 - \cancel{\sum Y_i^2} + \underline{110n \bar{Y}} - n 55^2 \right] \leq \log k$

$-\frac{1}{72} [10n \bar{Y} - n(55^2 - 50^2)] \leq \log k$

Reject H_0 if $10n\bar{Y} - n(55^2 - 50^2) \geq -72 \log k$

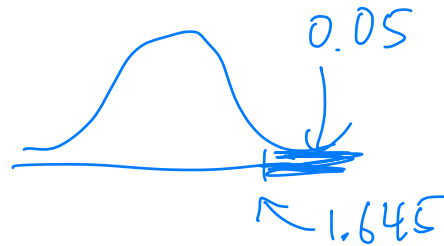
if $10n\bar{Y} \geq n(55^2 - 50^2) - 72 \log k$

if $\bar{Y} \geq \frac{n(55^2 - 50^2) - 72 \log k}{10n} = c$

Reject H_0 if
 $\bar{Y}$ "large"

$P(\bar{Y} \geq c) = 0.05$ if $\theta = 50$

$$P\left(\frac{\bar{Y} - 50}{6/\sqrt{n}} \geq \frac{c - 50}{6/\sqrt{n}}\right) = 0.05$$



$$P(Z > 1.645) = 0.05$$

$$\frac{c - 50}{6/\sqrt{n}} = 1.645$$

$$c - 50 = \frac{(1.645)6}{\sqrt{n}}$$

$$c = 50 + \frac{(1.645)(6)}{\sqrt{n}}$$

Reject H_0 if $\bar{Y} > 50 + \frac{1.645 \cdot 6}{\sqrt{n}}$

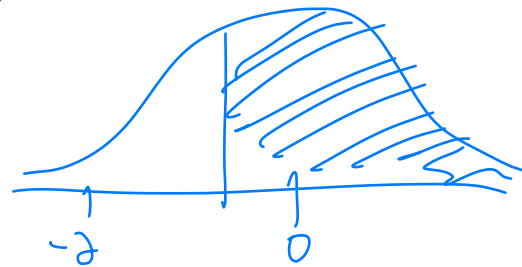
(b) power = $P(\text{Reject } H_0)$ if H_a is true $\Rightarrow \mu = 55$

$$c = 50 + \frac{(1.645)6}{\sqrt{9}} = 53.29$$

$$P(\bar{Y} > 53.29) \text{ if } \mu = 55$$

$$= P\left(\frac{\bar{Y} - 55}{6/\sqrt{9}} > \frac{53.29 - 55}{6/\sqrt{9}}\right)$$

$$= P(Z > -0.855)$$



$$1 - \text{pnorm}(-0.855)$$

0.8037